

## EXAM 7 – SPRING 2015

18. (2 points)

The International Actuarial Association's *A Global Framework for Insurer Solvency Assessment* outlines the following procedure for simulating underwriting losses for a multi-line insurer:

1. For each line of insurance  $i$ :
  - a. Select a random number  $\gamma_i$  from a gamma distribution with mean 1 and variance  $c_i$ .
  - b. Select a random claim count  $K_i$  from a Poisson distribution with mean  $\gamma_i \lambda_i$  where  $\lambda_i$  is the expected claim count for line of insurance  $i$ .
  - c. For each  $i$  and for  $k = 1, \dots, K_i$ , select a random claim size,  $Z_{ik}$ , from a lognormal distribution with mean  $\mu_i$  and standard deviation  $\sigma_i$ .
2. Set  $X_i = \sum_{k=1}^{K_i} Z_{ik}$  = Loss for line of insurance  $i$ .
3. Select a random number  $p$ , from a uniform (0, 1) distribution. For each line  $i$ , select  $\beta_i$  to be the  $p^{\text{th}}$  percentile of a distribution with  $E[\beta_i] = 1$  and  $\text{Var}[\beta_i] = b_i$ .
4. Set  $X = \sum_i \beta_i \cdot X_i$  = Loss for the insurer.

This results in the following formula for the variance of a line's loss ratio (losses divided by expected losses):

$$\text{Var} \left[ \frac{\beta_i \cdot X_i}{E[\beta_i \cdot X_i]} \right] = (1 + b_i) \left( \frac{\mu_i^2 + \sigma_i^2}{\lambda_i} + c_i \right) + b_i$$

a. (0.5 point)

Derive a formula for the minimum variance of a line's loss ratio for the model described above.

b. (0.25 point)

Briefly explain the significance of the result from part a. above for the estimation of an insurer's underwriting risk within any single line of business.

c. (0.25 point)

Identify the correlation affected by the parameter  $c_i$  above.

d. (0.25 point)

Identify the correlation affected by the parameter  $b_i$  above.

e. (0.75 point)

For a particular line of business, the standard deviation of the loss ratio can be no smaller than 30%. In addition, the standard deviation of inflationary effects is estimated at 4%. Determine estimates of  $b$  and  $c$  for this line of business.

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**EXAM 7 SPRING 2015 SAMPLE ANSWERS AND EXAMINERS' REPORT**

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| <b>QUESTION 18</b>                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                             |
| <b>TOTAL POINT VALUE: 2</b>                                                                                                                                                                                                                                                                                                                                                                                                       | <b>LEARNING OBJECTIVE:</b> C1: Demonstrate how insurance and financial risk can be analyzed quantitatively. |
| <b>SAMPLE ANSWERS</b>                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                             |
| <b>Part a: 0.5 point</b>                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                             |
| <p><i>Sample Answer 1</i><br/> The minimum variance occurs when the frequency <math>\rightarrow \infty</math> causing the process variance to go away</p> $\lim_{\lambda_i \rightarrow \infty} Var \left[ \frac{\beta X_i}{E[\beta_i X_i]} \right] = \lim_{\lambda_i \rightarrow \infty} \left[ (1 + b_i) \left( \frac{\mu_i^2 + \sigma_i^2}{\lambda_i^2} + c_i \right) + b_i \right] = (1 + b_i)c_i + b_i = c_i + b_i c_i + b_i$ |                                                                                                             |
| <p><i>Sample Answer 2</i><br/> Minimum variance when expected # of claims goes to infinity i.e. <math>\lambda_i \rightarrow \infty</math><br/> minimum ( <math>Var \left[ \frac{\beta X_i}{E[\beta_i X_i]} \right] ) = (1 + b_i)c_i + b_i</math></p>                                                                                                                                                                              |                                                                                                             |
| <b>Part b: 0.25 point</b>                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                             |
| <p><i>Sample Answer 1</i><br/> What this indicates is that regardless of size, even if when <math>\lambda \rightarrow \infty</math> and the maximum effect of the law of large numbers is achieved, there is always some variance that occur for a single line of business</p>                                                                                                                                                    |                                                                                                             |
| <p><i>Sample Answer 2</i><br/> If you increase the size of your book (increase <math>\lambda</math>) a lot, there is a minimum level of variance you will still have, cant lower it any more limit to diversification benefit</p>                                                                                                                                                                                                 |                                                                                                             |
| <p><i>Sample Answer 3</i><br/> lowest possible variance for this line of business</p>                                                                                                                                                                                                                                                                                                                                             |                                                                                                             |
| <b>Part c: 0.25 point</b>                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                             |
| <p><i>Sample Answer 1</i><br/> The parameter <math>c_i</math> represents the correlation between the claims within the given line(i), it determines the spread of claims within lines</p>                                                                                                                                                                                                                                         |                                                                                                             |
| <p><i>Sample Answer 2</i><br/> affects the correlation of losses within a LOB</p>                                                                                                                                                                                                                                                                                                                                                 |                                                                                                             |
| <p><i>Sample Answer 3</i><br/> c affects the mean for claim counts, if c is large the frequency will have larger variance</p>                                                                                                                                                                                                                                                                                                     |                                                                                                             |
| <b>Part d: 0.25 point</b>                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                             |
| <p><i>Sample Answer 1</i><br/> The correlation identified with <math>b_i</math> affects the correlation between line of business</p>                                                                                                                                                                                                                                                                                              |                                                                                                             |
| <p><i>Sample Answer 2</i><br/> Affects the correlation of losses between LOB</p>                                                                                                                                                                                                                                                                                                                                                  |                                                                                                             |
| <i>Sample Answer 3</i>                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                             |

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| <p>This is the correlation of the loss ratios by line</p> <p><i>Sample Answer 4</i><br/>b is correlated to the loss distribution, varying b will vary the percentile selected</p> <p><i>Sample Answer 5</i><br/>b correlates line l with the total company result, the larger the b the more correlated (inflationary effects)</p> <p><i>Sample Answer 6</i><br/>External system's correlation</p>                     |
| <b>Part e: 0.75 point</b>                                                                                                                                                                                                                                                                                                                                                                                              |
| <p><i>Sample Answer 1</i><br/> <math>\sqrt{b_i} = 0.04</math>; <math>b_i = 0.0016</math>;<br/> <math>Var = 0.3^2 = c_i + b_i + b_i c_i</math> ;<br/> <math>0.09 = c_i(1.0016) + 0.0016 \rightarrow c_i = 0.0883</math></p> <p><i>Sample Answer 2</i><br/> <math>0.3^2 = 0.04^2 + c_i + 0.04^2 c_i</math>    <math>b_i = 0.04^2 = 0.016</math>    <math>c_i = 0.0883</math></p>                                         |
| <b>EXAMINERS' REPORT</b>                                                                                                                                                                                                                                                                                                                                                                                               |
| <p>The key to this problem was to be able to interpret the formula provided: for example, parts a &amp; b required the candidates to understand that no matter how large <math>\lambda</math> is (meaning that the line is so large that there are many claim counts), there is a minimum to the variance of that line's loss ratio.</p> <p>Candidates typically earned little credit on this question as a whole.</p> |
| <b>Part a</b>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Only a few candidates seemed to know which parameter could be varied to derive the minimum variance.                                                                                                                                                                                                                                                                                                                   |
| <b>Part b</b>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Only a few candidates seemed to understand that the implication of the answer in part a was that the variance couldn't go below zero.                                                                                                                                                                                                                                                                                  |
| <b>Parts c &amp; d</b>                                                                                                                                                                                                                                                                                                                                                                                                 |
| Very few candidates could correctly interpret the role of $b_i$ and $c_i$ in the variance formula.                                                                                                                                                                                                                                                                                                                     |
| <b>Part e</b>                                                                                                                                                                                                                                                                                                                                                                                                          |
| Even among those candidates who could derive the minimum variance formula in part a, very few could use it to plug values in where provided to solve for the remaining quantities.                                                                                                                                                                                                                                     |